

Manish Dhakal

Computer Vision . Large Language Models

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Research

Efficient adaptation and knowledge compression of foundation models for resource-constrained deployment. Publications at premier venues, such as *EMNLP, WACV, MICCAI, ACCV, IJCNLP-AACL, CVPR-W* etc.

Education

Ph.D. in Industrial Engineering

University of Tennessee, Knoxville

Graduate School

August 2026 – May 2030*

Supervisor: [Yi Ding, Ph.D.](#)

- Research in *efficient adaptation* and *knowledge compression* of large pretrained vision, language, and multimodal models.
- Efficient *world models and VLAs* (since Aug. 2026); continuing *DistillLens* and *DistillHPC*.

Master's in Computer Science

Georgia State University

Graduate School

August 2024 – July 2026

- Admitted to the Ph.D. program in Computer Science; *earned the M.S. en route* before transferring to UTK with my advisor's lab in 2026.
- Deep learning projects on *2D/3D computer vision and large language models (LLMs)*, funded by the US Department of Defense (DoD).
- Started *DistillLens* (Oct. 2025) and *DistillHPC* (May 2026, with Oak Ridge National Laboratory).
- Grade: 3.96

Bachelor in Computer Engineering

Pulchowk Campus, Institute of Engineering, Tribhuvan University

Undergraduate School

November 2017 – April 2022

Thesis Supervisor: [Prof. Subarna Shakya](#)

- Ranked 11th (top 0.1%)* in the entrance exam, competing with *15,000+ candidates*, received a full scholarship for undergraduate studies.
- Gained knowledge about *significant CS courses* like AI, Image Processing, Data Structure & Algorithm, DBMS, Software Engineering, and so on.
- Thesis*: Automatic speech recognition for *low-resourced Nepali language* which was later presented at an IEEE conference.

Ongoing Projects

DistillLens

Knowledge Compression for LLMs

October 2025 – Present

- Aligns teacher and student intermediate layers via the *logit lens* with a *symmetric divergence*; outperforms standard distillation across GPT-2, Llama-2, and Qwen families with no added inference cost.

DistillHPC

Scientific Code Generation

with Oak Ridge National Laboratory

May 2026 – Present

- Label-free knowledge distillation* to train compact coding assistants for the parallel programming languages used on DOE supercomputers, without human-authored expert examples.

Efficient World Models and VLAs

Embodied AI

August 2026 – Present

- Applying *efficient adaptation* and *knowledge compression* to world models and vision-language-action (VLA) models, specializing one shared model to many tasks from limited demonstrations.

2022-Present

[Scholar](#) Citations: 140+ & h-index: 4 (*equal contribution)

- Budathoki, A., [Dhakal, M.](#), Ampel, B. M., & Ding, Y. (2026). Understanding the role of prompt template in knowledge distillation for safety alignment. In *EMNLP (Findings)*.
- Poudel, S., Kunwor, N., [Dhakal, M.](#), Jha, D., & Gaire, S. K. (2026). Federated LoRA adaptation of Biomed-CLIP across four international chest X-ray cohorts. In *DeCaF @ MICCAI*. [[Paper](#)]
- Poudel, S., Kunwor, N., Simkhada, R., Munir, M., [Dhakal, M.](#), & Poudel, K. (2026, June). Parameter efficient fine-tuning for domain-specific gastrointestinal disease recognition. In *PHAROS @ CVPR*. [[Paper](#), [Code](#)]
- [Dhakal, M.](#), Dasari, V. R., Sunderraman, R., & Ding, Y. (2026, March). GFT: Graph feature tuning for efficient point cloud analysis. In *WACV*. [[Paper](#), [Code](#)]
- [Dhakal, M.](#), Jinadu, U., Budathoki, A., Sunderraman, R., & Ding, Y. (2026, February). DistillLens: Symmetric knowledge distillation through logit lens. *arXiv preprint arXiv:2602.13567*. [[Paper](#), [Code](#)]
- Mim, S. T., Morris, J., [Dhakal, M.](#), Xiu, Y., Gorlatova, M., & Ding, Y. (2025, December). Can a unimodal language agent provide preferences to tune a multimodal vision-language model? In *IJCNLP-AACL Findings*. [[Paper](#)]
- Budathoki, A., & [Dhakal, M.](#) (2025). Adversarial robustness analysis of vision-language models in medical image segmentation. *arXiv preprint arXiv:2505.02971*. [[Paper](#), [Code](#)]
- Adhikari, R., Thapaliya, S., [Dhakal, M.](#), & Khanal, B. (2024). TuneVLSeg: Prompt tuning benchmark for vision-language segmentation models. In *ACCV (Oral)*. [[Paper](#), [Code](#)]
- [Dhakal, M.](#), Adhikari, R., Thapaliya, S., & Khanal, B. (2024). Finetuning vision-language segmentation efficiently with lightweight blocks. In *MICCAI*. [[Paper](#), [Code](#)]
- Adhikari, R.*, [Dhakal, M.*](#), Thapaliya, S.*, Poudel, K., Bhandari, P., & Khanal, B. (2023, October). Synthetic Boost: Leveraging synthetic data for enhanced vision-language segmentation in echocardiography. In *ASMUS @ MICCAI*. [[Paper](#), [Code](#)]
- Poudel, K.*, [Dhakal, M.*](#), Bhandari, P.*, Adhikari, R.*, Thapaliya, S.*, & Khanal, B. (2024). Exploring transfer learning in medical image segmentation using vision-language models. In *MIDL (Oral)*. [[Paper](#), [Code](#)]
- [Dhakal, M.](#), Chhetri, A., Gupta, A. K., Lamichhane, P., Pandey, S., & Shakya, S. (2022, July). Automatic speech recognition for the Nepali language using CNN, bidirectional LSTM, and ResNet. In *Proceedings of the International Conference on Inventive Computation Technologies (ICICT)*. IEEE. [[Paper](#), [Code](#)]

Work Experience

University of Tennessee, Knoxville

Graduate Research Assistant

Knoxville, TN, USA

August 2026 – May 2030*

Supervisor: [Yi Ding, Ph.D.](#)

- Research on efficient world models and VLAs; continuing *DistillLens* and *DistillHPC* (with ORNL).

Georgia State University

Graduate Research Assistant

Atlanta, GA, USA

August 2024 – July 2026

Supervisor: [Yi Ding, Ph.D.](#)

- Deep learning projects on *2D/3D computer vision* and *large language models (LLMs)*, funded by the US Department of Defense (DoD).
- Developed *parameter-efficient fine-tuning* for point clouds (GFT, WACV 2026) and *knowledge distillation* methods for LLMs (DistillLens; safety-alignment distillation, EMNLP Findings 2026).

Nepal Applied Mathematics and Informatics Institute for research (NAAMII)

Research Assistant

Lalitpur, Nepal

April 2022 – June 2024

Supervisor: [Bishesh Khanal, Ph.D.](#)

- Developed skills for *object detection and segmentation* tasks on 2D medical images and explored their multi-modal approach (esp. *vision-language models*); also worked on segmentation with *3D mesh* data.

- Demonstrated *strong skills in writing scientific manuscripts*, with multiple papers submitted for review, showcasing the ability to *communicate methodologies, results, and implications* effectively.
- Ensured *reproducibility and modularity in ML projects* by implementing robust methodologies and practices, allowing for the transparent and replicable programming of the projects.

Past Projects

Lower Limb Segmentation

July 2023 – September 2023

Medical Imaging

Supervisor: [Taman Upadhaya, Ph.D.](#)

- Conducted *training experiments* of different deep learning models on the remote server to segment three bones – knee, pelvis, and ankle – from CT scans of the lower limbs of patients.
- *Deployed a robust Python rest API* on the remote server for the segmentation request from a client, with a pipeline including pre-processing, inference, and post-processing steps.
- Ensured *interoperability, reproducibility, and understandability* of the deployed application using Docker, and well-structured documentation and comments.

Vision Language Segmentation Models (VLSMs) for Medical Images

May 2023 – August 2024

Medical Imaging

- Reported zero-shot and finetuned segmentation performance of *4 VLSMs* on *11 medical datasets* using *9 types of prompts* derived from *14 attributes*, prompts are given as text conditioning information.
- Worked with *encoder-decoder architecture* to generate binary segmentation masks for VLSMs.
- Tested the compatibility of the VLSMs (such as *CLIPSeg and CRIS*) pre-trained for open-domain images with medical images.

Object Detection in 2D Orthopantomogram (OPG) Images

September 2022 – August 2024

Dental Imaging

- Critically analyzed the *literature and state-of-the-art* models for different segmentation and detection tasks on radiology images of dentistry and their inadequacy.
- Designed and developed the *data annotation tool* for object detection over 2D OPG images.
- Working on identification and localization of dental *anatomical structures and abnormalities* while benchmarking with existing methods like *YOLO, RetinaNet, RCNN, and FastRCNN*.

Segmentation in 3D Teeth Scan

Summer 2022

MICCAI Challenge 2022

- Learned about the representation and preprocessing of *3D mesh and point cloud* data.
- Benchmarked with different 3D point cloud segmentation models such as *Pointnet++ and DeltaConv*.

Nepali AutoComplete and LM

August 2020 – October 2020

[Open Source Project](#)

- Designed and trained *language model of Nepali (ie. Devnagari transcript)* for the text auto-complete system.
- Programmed the *pre-processing pipeline* to remove the non-Nepali characters from the dataset.

Super-Resolution with GAN (SRGAN)

May 2020 – August 2020

[Open Source Project](#)

- Implemented *open source model* of *SRGAN* with Keras/TensorFlow.
- Developed the *understanding of generator and discriminator* in GAN-based generative models.

Teaching

Community Eye, ENT & Rehabilitation Center (CEERS)

Trainer

Bhaktapur, Nepal

June 2023 – June 2024

- *Training a group of interns* to develop medical imaging applications with the use of ML.
- Instructing and guiding them about ML through activities like *paper reading sessions*, *lecture-lab sessions*, and *topic presentations*.

4th Annual Nepal AI School (AN AIS)

Teaching Assistant

Kathmandu, Nepal

May 2023 – June 2023

- *Guided participants* through a series of labs related to *neural networks*, *transformers*, *federated learning*, *graph neural networks*, *active learning*, and so on.
- *Mentored three groups* during the 10-day *machine learning hackathon* (namely, Hack-a-Dev).

Software Fellowship, Locus 2021

Programming Instructor

Online

Summer 2021

- Provided tutoring on *software development life cycle* and assisted participants with *software documentation* and *library/framework installation*.
- Taught participants about *API development for web applications*, emphasizing its concepts, best practices, and usage.

Achievements/Services

- 2026** *MICCAI Reviewer* (invited; 4 papers reviewed).
- 2025** *AAAI Reviewer* (invited; 8 papers reviewed).
- 2024** *Presidential Fellowship*, GSU (Only 6% of the doctoral students @ GSU).
- 2024** *LMIC Travel Grant* by the conference of MICCAI.
- 2017** *Undergraduate* funded by the Government of Nepal.

Technical skills

- Deep Learning** *Unimodal and multimodal learning*, large language models (LLMs), model quantization, efficient learning, reinforcement learning fine-tuning, 2D/3D vision, convolution, and transformers.
- Writing** *Knowledge synthesis* from the existing literature, *writing scientific documents* and *manuscripts* with *LaTeX*, and *communicating the results* to the community with transparency.
- Remote Server** Able to work with *remote Linux machines* for coding and project deployment using *SSH*, *shell script*, *tmux*, *Nginx*, and *Docker*.

References (Research Advisors)

Yi Ding, Ph.D.

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Raj Sunderraman, Ph.D.

Professor, Department of Computer Science, Georgia State University (GSU)
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Bishesh Khanal, Ph.D.

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